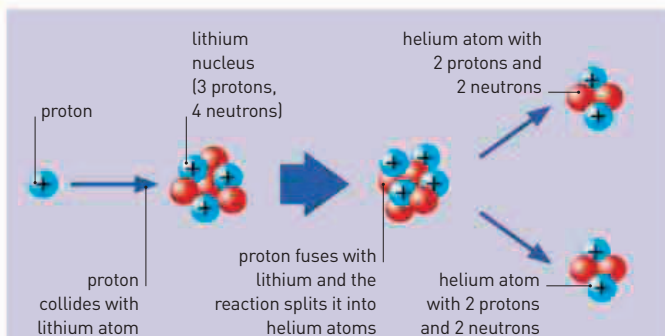


22.5 PERCENT

THE PROPORTION OF THE UNIVERSE THAT IS DARK MATTER



SPLITTING THE ATOM

Transmutation of elements was first achieved by changing nitrogen to oxygen (see 1917). In 1932, John Cockcroft and Ernest Walton used a similar technique to split lithium. Lithium atoms are the lightest of any metal and contain just three protons. When an extra proton collides with an atom, a nuclear reaction occurs: the four-neutron, four-proton total splits into two helium atoms.

In 1932, **Hans Krebs** (1900–81), a German-born biochemist, was looking at how the body processes waste nitrogen. Excess amino acids (the building blocks of protein) are recycled into carbohydrate. Krebs discovered the **cycle of chemical reactions** used by **liver cells** to process the **nitrogen content** into a compound called **urea**, which is excreted by the body.

Dutch and Swiss astronomers **Jan Oort** (1900–81) and **Fritz Zwicky** (1898–1974) discovered an **anomaly in galaxies**: they were much larger than the quantity of star material

suggested. This indicated the existence of a type of matter not previously detected. It became known as **dark matter**—because it neither emits nor absorbs light.

In July 1933, Polish-born chemist **Tadeus Reichstein** (1897–1996) became the first person to make a vitamin by artificial means when he made ascorbic acid (**vitamin C**).

Positron–electron tracks

Charged particles appear to shoot upwards and shower down, forming tracks. In a magnetic field, negative electrons coil one way and positive particles (positrons) the other.



February 27, 1932 James Chadwick describes evidence for the existence of the neutron

August 12, 1932 Carl Anderson discovers the positive electron-like particles later called positrons

1932 British biologist John B.S. Haldane publishes *Causes of Evolution*, uniting genetics and evolutionary theory

February 16, 1933 Fritz Zwicky provides evidence for dark matter

February 24, 1933 Vladimir Zworykin describes principles of electron microscopy

September 11, 1933 New Zealand-born physicist Ernest Rutherford says that there is no future in nuclear energy

April 14, 1932 John Cockcroft and Ernest Walton split the atomic nucleus of lithium

August 17, 1932 Jan Oort gives evidence for existence of dark matter

February 14, 1933 French astronomer Ernest Esclangon starts first talking clock telephone service

July 11, 1933 Tadeus Reichstein reports making vitamin C artificially

September 17, 1933 German physicist Albert Einstein arrives in the UK after fleeing Nazi Germany; he reaches America in October